



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

06-04-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE PHASE-I to VIII (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4				
	A.	A,C,D	B,D	A,B,D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.50	4.00	0.16	6.00	0.50	36.00	0.50	2.00
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	3	4	8	6	7	1	3

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4				
	A.	A,C	A,C	B,C	A,C,D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.70	60.00	5.50	11.00	5.00	1.25	1.06	10.00
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	5	5	1	3	7	2	3	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4				
	A.	A,D	A	B,C,D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	664.00	288.00	9.00	1.00	35.00	24.00	0.12 or 0.13	13.00
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	3	1	2	2	1	0	4

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (A,C,D)**

 Velocity of efflux : $v = \sqrt{2gh}$

$$\frac{dv}{dt} = A \frac{dh}{dt} = -av = -a\sqrt{2gh}$$

$$\frac{dh}{dt} = -\frac{a}{A} \sqrt{2gh}$$

$$\int_{\ell}^h \frac{dh}{\sqrt{h}} = -\frac{a}{A} \sqrt{2g} \int_0^t dt$$

$$2(\sqrt{h} - \sqrt{\ell}) = -\frac{a}{A} \sqrt{2g} t$$

$$\sqrt{h} = \sqrt{\ell} - \frac{a}{\sqrt{2A}} \sqrt{g} t$$

$$\frac{dh}{dt} = -\frac{a}{A} \left(\sqrt{2g\ell} - \frac{a}{A} gt \right)$$

$$\left| \frac{dh}{dt} \right| = \frac{a}{A} \left(\sqrt{2g\ell} - \frac{a}{A} gt \right)$$

$$C_{eq} = \frac{k\epsilon_0 h \ell}{d} + \frac{\epsilon_0 (\ell - h) \ell}{d}$$

$$q = CE$$

$$\frac{dq}{dt} = i = E \frac{dc}{dt} = \frac{\epsilon_0 \ell E}{d} (k-1) \frac{dh}{dt}$$

2. Ans (B,D)

Velocity of efflux : $v = \sqrt{2gh}$

$$\frac{dv}{dt} = A \frac{dh}{dt} = -av = -a\sqrt{2gh}$$

$$\frac{dh}{dt} = -\frac{a}{A}\sqrt{2gh}$$

$$\int_{\ell}^h \frac{dh}{\sqrt{h}} = -\frac{a}{A}\sqrt{2g} \int_0^t dt$$

$$2(\sqrt{h} - \sqrt{\ell}) = -\frac{a}{A}\sqrt{2g}t$$

$$\sqrt{h} = \sqrt{\ell} - \frac{a}{\sqrt{2}A}\sqrt{g}t$$

$$\frac{dh}{dt} = -\frac{a}{A}\left(\sqrt{2g\ell} - \frac{a}{A}gt\right)$$

$$\left|\frac{dh}{dt}\right| = \frac{a}{A}\left(\sqrt{2g\ell} - \frac{a}{A}gt\right)$$

$$C_{eq} = \frac{k\epsilon_0 h\ell}{d} + \frac{\epsilon_0 (\ell - h)\ell}{d}$$

$$q = CE$$

$$\frac{dq}{dt} = i = E \frac{dc}{dt} = \frac{\epsilon_0 \ell E}{d}(k-1) \frac{dh}{dt}$$

3. Ans (A,B,D)

$$mg = \mu \frac{mv^2}{r} \Rightarrow 1600 = \mu \times 160 \times \frac{100}{10}$$

$$\Rightarrow \mu = 1$$

4. Ans (C)

$$N_{bot} = mg = \frac{mv^2}{R}$$

$$N_{bot} = 3200 \text{ N}$$

PART-1 : PHYSICS

SECTION-II

2. Ans (4.00)

$$u = \int_0^2 Li \, di$$

$$= \frac{L}{2} [i^2]_0^2 = 4J$$

3. Ans (0.16)

$$f = \frac{V}{2(\ell + 2e)} \quad e = 0.6r$$

$$= \frac{V}{2(\ell + 1.2r)}$$

$$\frac{\Delta f}{f} = \frac{\Delta V}{V} - \frac{\Delta(\ell + 1.2V)}{\ell + 1.2r}$$

$$= -\left(\frac{\Delta\ell + 1.2\Delta V}{\ell + 1.2r}\right)$$

$$\frac{\Delta f}{f} \times 100 = \left(\frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5}\right) \times 100$$

$$= 0.16\%$$

4. Ans (6.00)

By Gauss Law

$$\int \vec{E} \cdot d\vec{A} = \frac{\sigma Q}{\epsilon_0}$$

$$\frac{\sigma r}{\epsilon_0} (4\pi r^2) = \frac{\sigma Q}{\epsilon_0}$$

$$20r^3 = \sigma Q$$

$$20\left(\frac{3}{10}\right) = \sigma Q$$

$$\sigma Q = 6$$

6. Ans (36.00)

$$\frac{v}{u} = 3$$

$$\frac{1}{3u} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{f} = \frac{4}{3u} \quad (1)$$

Ind case

$$\frac{v}{(u+6)} = 2$$

[u already is negative]

$$\frac{1}{2(u+6)} + \frac{1}{u+6} = \frac{1}{f} \quad (2)$$

from (1) & (2)

$$u = 48$$

$$f = 36$$

$$v_1 = 144$$

$$v_2 = 108$$

$$\text{shift} = v_1 - v_2$$

$$= 36$$

7. **Ans (0.50)**

Conservation of mechanical energy

$$\frac{-GMm}{2R} + \frac{1}{2} \frac{mGM}{2R} + x = 0$$

$$x = \frac{mgR}{4}$$

8. **Ans (2.00)**

$$\frac{V_{\text{output}}}{V_{\text{input}}} = \frac{1}{5}$$

$$\Rightarrow V_{\text{output}} = \frac{100}{5} = 20 \text{ volt}$$

$$\therefore I = \frac{20}{10} = 2A$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (3)**



$P = 2\sigma AT_1^4 - \sigma AT_2^4$ (steady-state condition of sheet I)(i)

and $2\sigma AT_2^4 = \sigma AT_1^4$ (steady state condition of sheet II)(ii)

$$\Rightarrow P = 3\sigma AT_2^4 = 3kW$$

2. **Ans (3)**

CA is adiabatic, BC is isothermal as per slope.

$$\text{Hence } V_1 = \frac{P_0 V_0}{3P_0} = \frac{V_0}{3}$$

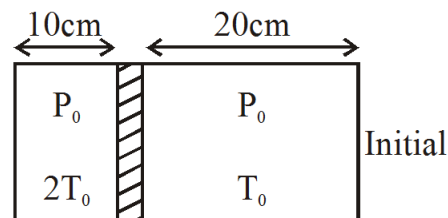
$$\text{Using } PV^\gamma = \text{const} \Rightarrow \gamma = \frac{\ln 6}{\ln 3}$$

3. **Ans (4)**

For chamber (1)

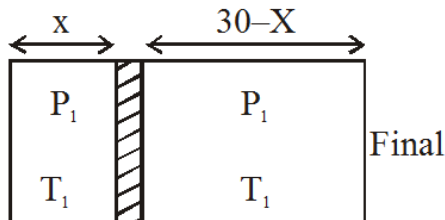
equation number of moles

$$\frac{P_0(10)(A)}{R(2T_0)} = \frac{P_1 x (A)}{RT_1} \dots\dots(1)$$



For chamber (2)

$$\frac{P_0(20)(A)}{T_0} = \frac{P_1(30-x)A}{RT_1} \dots\dots(2)$$



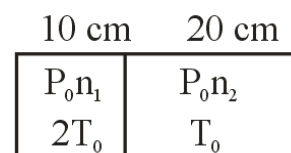
Dividing (1) by (2)

$$\frac{1}{4} = \frac{x}{30-x} \Rightarrow 30-x = 4x$$

$$5x = 30$$

thus piston moves by 4cm

OR



$$P_0 A(10) = n_1(2T_0)R$$

$$n_1 = 5 \left(\frac{P_0 A}{T_0 R} \right)$$

$$P_0 A(20) = n_2 T_0 R$$

$$n_2 = 20 \left(\frac{P_0 A}{T_0 R} \right)$$

At equilibrium

$$T_f \Rightarrow T_1 = T_2$$

$$P_1 = P_2$$

$$\frac{10-x}{20+x} = \frac{n_1}{n_2} \text{ (x is displacement of piston)}$$

$$\frac{10-x}{20+x} = \frac{1}{4}$$

$$\therefore x = 4 \text{ cm}$$

4. Ans (8)

Total number of revolution = $\frac{\ell}{\frac{v \cos 30^\circ}{T}}$ where

$$T = \frac{2\pi m}{qB} \text{ and } B = \mu_0 n i = \frac{\mu_0 i N}{A}$$

5. Ans (6)

$$F = \frac{dN}{dt} \times mv$$

$$= \frac{\rho \lambda}{hc} \times \sqrt{2mk}$$

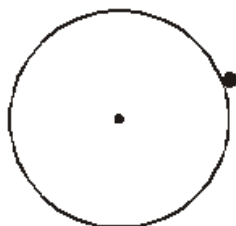
$$= \frac{\rho \lambda}{hc} \times \sqrt{2mqv}$$

$$= \frac{100}{6.25 \times 1.6 \times 10^{-19}} \sqrt{2 \times 9 \times 10^{-31} \times 1.6 \times 10^{-19} \times 125}$$

$$1 \times 10^{20} \times 10^{-25} \times \sqrt{25 \times 9 \times 16}$$

$$= 60 \times 10^{-5} = 6 \times 10^{-4} N$$

6. Ans (7)



$$r = R_0 A^{1/3} = 1.2 \times 10^{-15} \times 4 = 4.8 \times 10^{-15}$$

$$F = \frac{9 \times 10^9 \times 28 \times (1.6 \times 10^{-19}) \times 1.6 \times 10^{-19}}{4.8 \times 10^{-15} \times 4.8 \times 10^{-15}}$$

$$= 280 N$$

$$\frac{F}{40} = 7$$

7. Ans (1)

A B

N N

After 2 half life of B

$$\frac{N}{2^n} = \left(\frac{N}{2^2} \right) \frac{1}{2}$$

$$n = 3$$

3 half life of A has passed in 3 day $t_{1/2} A = 1$

day

8. Ans (3)

For $A \leq 50$; binding energy is given as

$$E = \frac{3A^2}{25} + 2A$$

$$A_{45} + A_{30} \rightarrow A_{75} + Q$$

If energy released is positive reaction will be spontaneous

$$Q = \frac{3}{25} (75^2 - 30^2 - 40^2) + 2(75 - 30 - 40)$$

$$Q > 0$$

Similarly $A_{10} + A_{20} \rightarrow A_{30} + \theta$, releases

energy and $A_{80} \rightarrow 2A_{40}$; do not releases energy.

For $A > 100$

$$E = 12A - \frac{A^2}{25} \text{ is the binding E of a nuclear}$$

$$A_{150} \rightarrow 2A_{75} + Q$$

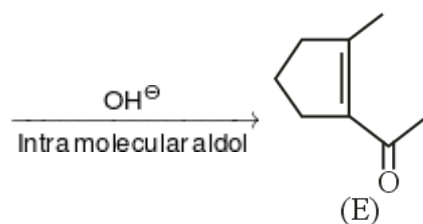
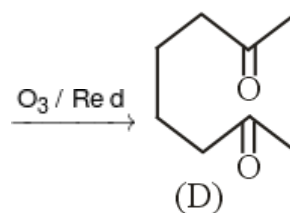
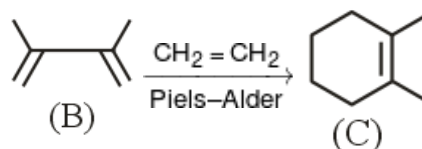
$$Q = [2(8)(75)] - \left[12(150) - \frac{(150)^2}{25} \right]$$

$$Q > 0$$

PART-2 : CHEMISTRY

SECTION-I

2. Ans (A,C)



4. **Ans (A,C,D)**



PART-2 : CHEMISTRY

SECTION-II

4. **Ans (11.00)**

$$x = 8$$

$$y = 2$$

$$z = 1$$

5. **Ans (5.00)**

$$\pi = CRT = 0.4 \text{ M} \times 0.0821 \frac{\text{L. atm}}{\text{molK}} \times 300 \text{ K}$$

$$= 9.852 \text{ atm} \approx 10 \text{ atm.}$$

6. **Ans (1.25)**

$$\rho = \frac{ZM}{N_A a^3}$$

$$M = \frac{\rho \times N_A \times a^3}{Z}$$

$$= \frac{0.8 \times 6 \times 10^{23} \times 512 \times 10^{-24}}{4} = 61.44 \text{ g}$$

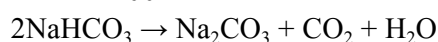
$$\therefore \text{Number of moles of 'M'} = \frac{307.2}{61.44} = 5$$

$$\text{Number of moles of unit cell} = \frac{5}{4} = 1.25$$

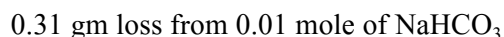
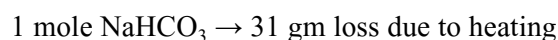
7. **Ans (1.06)**

Loss in wt. due to heating

$$= 6.2 \times \frac{5}{100} = 0.31 \text{ g}$$



$$2 \times 84 \text{ gm} \quad 44 + 18 \text{ gm loss}$$



$$\text{Moles of Na}_2\text{CO}_3 \text{ produced} = \frac{0.01}{2} = 0.005$$

Total moles of carbonate reacted with $\text{BaCl}_2 =$

$$\left(7.5 \times 0.2 \times \frac{100}{10} \right) \times 10^{-3} = 0.15$$

Moles of carbonates in original sample = 0.015

$$- 0.005 = 0.01$$

Mass of Na_2CO_3 in original sample = $0.01 \times$

$$106 = 1.06 \text{ gram}$$

8. **Ans (10.00)**

Let initial state

Final state (Isobaric)

Final state (Isothermal)

$$P \quad \rightarrow \quad P$$

$$P/2$$

$$V \quad \rightarrow \quad 2V$$

$$2V$$

$$W_{\text{isobaric}} = -P[2V - V]$$

$$= -PV = -20 \text{ kJ}$$

$$PV = 20$$

$$W_{\text{isothermal}} = -P_{\text{ext}} [2V - V]$$

$$= -\frac{P}{2} \times V = -\frac{20}{2} = -10 \text{ kJ}$$

PART-2 : CHEMISTRY

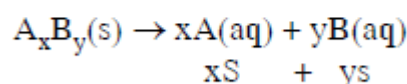
SECTION-III

7. **Ans (3)**

$$m = \frac{3.12 \times 10^{-3}}{0.52} = 6 \times 10^{-3} \text{ mole / kg}$$

$$\cong \text{molarity} \cong 6 \times 10^{-3} \text{ mole / L}$$

$$S = 2 \times 10^{-3} \text{ mole/L}$$



$$\text{total molarity} = (x \times y)s = 6 \times 10^{-3}$$

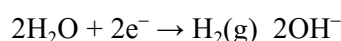
$$x + y = 3$$

8. **Ans (4)**

$$[\text{H}^+] = 10^{-9} \text{ M}$$

$$[\text{OH}^-] = 10^{-5} \text{ M}$$

$$n_{\text{OH}^-} = 10^{-5} \times 0.8$$



$$\frac{0.193 \times t}{96500} \times 8 \times 10^{-6}$$

$$t = 4 \text{ sec.}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,D)**

$$A(-2\sqrt{3}, 2)$$

$$B(2\sqrt{3}, 2)$$

$$C(2, 2\sqrt{3})$$

$$D(2, -2\sqrt{3})$$

Suppose $T_1(\alpha_1, \beta_1)$

$$T_2(\alpha_2, \beta_2)$$

$$T_3(\alpha_3, \beta_3)$$

chord of contact of tangents drawn from T_1

$$\Rightarrow x\alpha_1 + y\beta_1 = 16 \quad \dots (1)$$

equation of AB

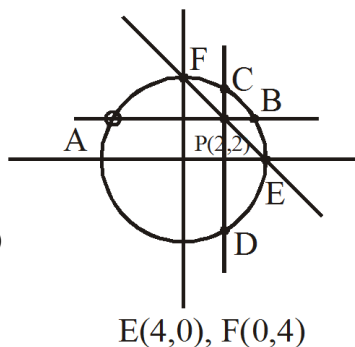
$$\Rightarrow y = 2 \quad \dots (2)$$

(1) & (2) both are representing the same line

$$\Rightarrow T_1(0, 8)$$

Similarly $T_2(8, 0)$

$$T_3(4, 4)$$



2. **Ans (A)**

$$P(4\cos\theta, 4\sin\theta)$$

tangent at P

$$x\cos\theta + y\sin\theta = 4$$

$$A\left(\frac{4}{\cos\theta}, 0\right),$$

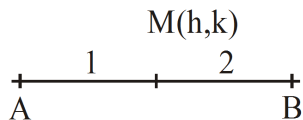
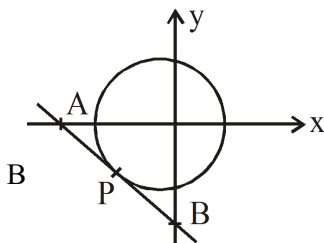
$$\left(0, \frac{4}{\sin\theta}\right)$$

$$\frac{8}{\cos\theta \times 3} = h \Rightarrow \cos\theta = \frac{8}{3h}$$

$$\frac{4}{\sin\theta \times 3} = k \Rightarrow \sin\theta = \frac{4}{3k}$$

$$\Rightarrow \frac{64}{9x^2} + \frac{16}{9y^2} = 1$$

$$\Rightarrow 4y^2 + x^2 = \frac{9x^2y^2}{16}$$



3. **Ans (B,C,D)**

Required probability

$$= \frac{4! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) + 2}{6!} = \frac{11}{720}$$

4. **Ans (B)**

$$S_1S_2 \quad S_3S_4 \quad S_5S_6$$

Required probability

$$= \frac{3! \times 2 \times 2 \times 2}{6!}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (664.00)**

Number must have

$$(12), (16), (24), (32), (36), (44), (52), (56), (64)$$

as last 2 digits

$\therefore$ Total number of numbers

$$= 6 \times 6 \times 6 \times 6 \times 9 = 6^4 \times 9$$

2. **Ans (288.00)**

$$X : 2, 7, 12, \dots, 10087, 10092$$

$$Y : 3, 10, 17, \dots, 14122, 14129$$

$$X \cap Y : 17, 52, 87, \dots$$

$$\text{Let } m = n(X \cap Y)$$

$$\therefore 17 + (m-1) \times 35 \leq 10092$$

$$\Rightarrow m = 288$$

3. **Ans (9.00)**

$$\left((\log_3 8)^3 \right)^{\frac{1}{\log_3 (\log_3 8)}} = (\log_3 8)^{3 \log_3 (\log_3 8)} = 3^3$$

$$(\sqrt{5})^{\frac{1}{\log_9 5}} = (\sqrt{5})^{\log_{\sqrt{5}} 3} = 3$$

$$\text{Required no} = \frac{3^3}{3} = 9$$

4. **Ans (1.00)**

$$f(x) = \frac{x^2}{1-x} - \frac{x^2}{2-x}; g(x) = \frac{-x^2}{2+x} + \frac{x}{1+x}$$

$$f(x) = g(x)$$

$$\Rightarrow x^3 + 2x^2 + 5x - 2 = 0 \text{ or } x = 0$$

5. **Ans (35.00)**

$$m = {}^{10}C_1 \times 9! = 10 \times 9!$$

$$n = {}^9C_1 \times \frac{10!}{2!} = 45 \times 9!$$

$$n - m = 35 \times 9!$$

$$\Rightarrow \lambda = 35$$

6. **Ans (24.00)**

$$P(\alpha, \beta, \gamma)$$

$$Q(\alpha, 0, 0)$$

$$R(-\alpha, \beta, \gamma)$$

PQ is perpendicular to $y + z = 5$

$$\Rightarrow \beta = \gamma$$

Also mid point of PQ lies in $y + z = 5$

$$\Rightarrow \beta + \gamma = 10 \Rightarrow \beta = \gamma = 5$$

Now distance of point P

from y-axis = 13

$$\Rightarrow \alpha^2 + \gamma^2 = 169$$

$$\Rightarrow \alpha^2 = 12^2$$

$$\Rightarrow \alpha = 12$$

Now PR = 24

7. **Ans (0.12 or 0.13)**

A(10,0,0), B(0,10,0), C(0,0,10), D(5,5,5),

E(10,10,10)

$$\vec{\alpha} = 5\hat{i} - 5\hat{j} - 5\hat{k}$$

$$\vec{\beta} = -5\hat{i} + 5\hat{j} - 5\hat{k}$$

$$\vec{\gamma} = -5\hat{i} - 5\hat{j} + 5\hat{k}$$

$$\vec{\delta} = 5\hat{i} + 5\hat{j} + 5\hat{k}$$

$$\left| (\vec{\alpha} \times \vec{\beta}) \times (\vec{\gamma} \times \vec{\delta}) \right| = 25 \times 25 \times 8$$

8. **Ans (13.00)**

$$X = \sum_{r=0}^{10} (2r+3) \binom{10}{r}^2$$

$$= 2 \times \sum_{r=0}^{10} r \cdot \binom{10}{r} \cdot \binom{10}{r} + 3 \sum_{r=0}^{10} \binom{10}{r}^2$$

$$= 2 \sum_{r=0}^{10} 10 \cdot {}^9C_{r-1} \cdot {}^{10}C_r + 3 \times {}^{20}C_{10}$$

$$= 20 \sum_{r=0}^{10} {}^{10}C_r \cdot {}^9C_{10-r} + 3 \times {}^{20}C_{10}$$

$$= 20 \times {}^{19}C_{10} + 3 \times {}^{20}C_{10}$$

$$= 13 \times {}^{20}C_{10}$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (1)**

$$\vec{c} = \alpha \vec{a} + \beta \vec{b} + \frac{(\vec{a} \times \vec{b})}{2019}$$

$$\vec{c} \cdot \vec{a} = \alpha a^2 \Rightarrow \alpha = \sqrt{2} \cos \theta$$

$$\vec{c} \cdot \vec{a} = \beta b^2 \Rightarrow \beta = \sqrt{2} \cos \theta$$

$$|\vec{a} \times \vec{b}| = (2019)^4$$

$$\vec{c} \cdot \vec{c} = \alpha^2 a^2 + \beta^2 b^2 + (2019)^2$$

$$\Rightarrow 2 \times (2019)^2 = (2019)^2 2\cos^2 \theta + (2019)^2$$

$$2\cos^2 \theta + (2019)^2$$

$$\Rightarrow 2 = 2\cos^2 \theta + 2\cos^2 \theta + 1$$

$$\Rightarrow 4\cos^2 \theta = 1$$

2. **Ans (3)**

$$\sqrt{3}a \cos \alpha + 2b \sin \alpha = 2019$$

$$\sqrt{3}a \cos \beta + 2b \sin \beta = 2019$$

$$\sqrt{3}a (\cos \alpha - \cos \beta) + 2b (\sin \alpha - \sin \beta) = 0$$

$$\sqrt{3}a \times 2 \sin \frac{\alpha+\beta}{2} \cdot \sin \frac{\beta-\alpha}{2} + 2b \times 2 \cos \frac{\alpha+\beta}{2} \cdot \sin \frac{\alpha-\beta}{2} = 0$$

$$2\sqrt{3}a \sin \frac{\alpha+\beta}{2} - 4b \cos \frac{\alpha+\beta}{2} = 0$$

$$2\sqrt{3}a \times \frac{1}{2} - 4b \times \frac{\sqrt{3}}{2} = 0$$

$$\Rightarrow a = 2b$$

3. **Ans (1)**

$$\text{Area taken away} = \int_0^1 (x - x^3) dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\Rightarrow \text{Area of the actual land} = 1$$

$$\Rightarrow \frac{1}{2} (a^2 + 1) \times 1 = 1$$

$$\Rightarrow a^2 = 1$$

4. Ans (2)

$$2\sqrt{2} \cos 15^\circ = 4 \sin 45^\circ \cos 15^\circ$$

$$= 2 (\sin 60^\circ + \sin 30^\circ)$$

$$= (\sqrt{3} + 1)$$

$$\text{Now } \int \frac{2 \cos x \, dx}{((\sin x + 1)^{1010} (1 - \sin x)^{3030})^{\frac{1}{2020}}}$$

$$= \int \frac{2 \cos x \, dx}{(1 + \sin x)^2 \left(\frac{1 - \sin x}{1 + \sin x}\right)^{3/2}}$$

$$\text{Put } \frac{1 - \sin x}{1 + \sin x} = t$$

$$\frac{(-\cos x (1 + \sin x) - (1 - \sin x) \cos x)}{(1 + \sin x)^2} dx = dt$$

$$\Rightarrow \frac{2 \cos x \, dx}{(1 + \sin x)^2} = -dt$$

$$= - \int \frac{dt}{t^{3/2}} = \frac{2}{\sqrt{t}} + c$$

$$\text{So, } \int_0^{\pi/6} \frac{(2\sqrt{2} \cos 15^\circ) \cos x \, dx}{((\sin x + 1)^{1010} (1 - \sin x)^{3030})^{\frac{1}{2020}}}$$

$$= (1 + \sqrt{3})(\sqrt{3} - 1) = 2$$

5. Ans (2)

$$\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 -$$

$$(a_3 b_2 c_1 + b_3 c_2 a_1 + a_2 b_1 c_3)$$

when 0, 0, 0, 0, 1, 1, 1, 1, -1 are used

maximum value of $\Delta = 2$

when 0, 0, 0, 0, 0, 1, 1, 1, -1 maximum value

of $\Delta = 1$

when 0, 0, 0, 0, 0, 0, 1, 1, +1 maximum value

of $\Delta = 1$

when number of zeros is more than 6,

maximum value of $\Delta = 0$

6. Ans (1)

$$L = \lim_{n \rightarrow \infty} \left(\frac{(n^2 + 1^2)(n^2 + 2^2)(n^2 + 3^2) \dots (n^2 + n^2)}{(n^2)^n} \right)^{\frac{1}{2n}}$$

$$= \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(1 + \frac{r^2}{n^2} \right)^{\frac{1}{2n}}$$

$$\Rightarrow \log_e L = \lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{r=1}^n \ln \left(1 + \left(\frac{r}{n} \right)^2 \right)$$

$$= \frac{1}{2} \int_0^1 \ln(1 + x^2) \, dx$$

$$= \lambda$$

$$\text{where } 0 < \lambda < \frac{\ln 2}{2}$$

$$\Rightarrow [L] = [e^\lambda] = 1$$

7. Ans (0)

$$\frac{dy}{(2019)^2 - (2020)^2 y^2} = dx$$

$$\frac{1}{2 \times \left(\frac{2019}{2020} \right) \times (2020)^2} = x + c$$

$$n \left| \left(\frac{2019 + 2020y}{2019 - 2020y} \right) \right|$$

$$\Rightarrow \left| \frac{2019 + 2020y}{2019 - 2020y} \right| = e^{(2 \times 2019 \times 2020)x + c}$$

Taking limit $x \rightarrow -\infty$

$$\Rightarrow \lim_{x \rightarrow -\infty} \left| \frac{2020f(x) + 2019}{2019 - 2020f(x)} \right| = 0$$

$$\Rightarrow \lim_{x \rightarrow -\infty} 2020 f(x) = -2019$$

$$\Rightarrow \lim_{x \rightarrow \infty} f(-x) = \frac{-2019}{2020}$$

$$\Rightarrow [[\lambda]] = 0$$

8. Ans (4)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) + f(h) + 2xh - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) + 2xh}{h}$$

$$= f'(0) + 2x$$

$$= 2x$$

$$\Rightarrow f(x) = x^2$$

$$\Rightarrow f(2) = 4$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE PHASE-I to VIII (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	B	C	B	D	A	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	D	A	B	A	C	B	D	D

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	D	B	C	D	B	A	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	D	C	D	C	D	B	B	D

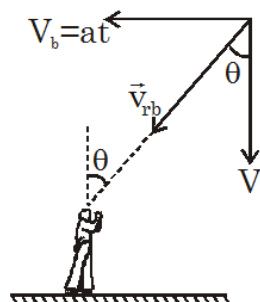
PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	B	B	A	B	B	A	B	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	B	D	D	D	D	A	C

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (A)**

$$\vec{v}_{r,b} = \vec{v}_r - \vec{v}_b$$



$$\tan \theta = \frac{at}{v} = t$$

$$\Rightarrow \sec^2 \theta \frac{d\theta}{dt} = 1$$

$$\frac{d\theta}{dt} = \frac{1}{\sec^2 \theta} = \frac{1}{(1-t^2)}$$

 2. **Ans (A)**

$$\frac{x}{2} = \frac{\ell}{1-\ell}$$

$$\frac{2}{x} = \frac{\ell+0.2}{1-(\ell+0.2)} - \frac{\ell-0.2}{0.8-\ell}$$

$$\frac{\ell(\ell+0.2)}{(1-\ell)(0.8-\ell)} - 1$$

$$\ell^2 + 0.2\ell = 0.8 - \ell^2 - 0.8\ell - 1$$

$$2\ell = 0.8$$

$$\ell = 0.4 \text{ m}$$

$$\frac{x}{2} = \frac{0.4}{0.6} \Rightarrow x = \frac{4}{3}$$

$$\frac{2}{x} = \frac{\ell-0.2}{1-(\ell-0.2)} = \frac{\ell-0.2}{1.2-\ell}$$

$$1 = \frac{\ell(\ell-0.2)}{(1-\ell)(1.2-\ell)}$$

$$\ell^2 = 0.2\ell - 1.2 + \ell^2 + 2.2\ell$$

$$2\ell = 1.2$$

$$\ell = 0.6$$

$$\Rightarrow x = \frac{2 \times 0.6}{0.4} = 3\Omega$$

3. **Ans (B)**

Using Mosely's law for both cobalt and impurity

$$\sqrt{\nu} = K(Z - 1)$$

$$\Rightarrow \sqrt{\frac{c}{\lambda_{CO}}} = K(Z_{CO} - 1) \text{ and } \sqrt{\frac{c}{\lambda_x}} = k(Z_x - 1)$$

$$\sqrt{\frac{\lambda_{CO}}{\lambda_x}} = \frac{Z_x - 1}{Z_{CO} - 1} \Rightarrow Z_x = 40$$

4. **Ans (B)**

given that, perimeter is decreasing at a rate of $2v$

$$\therefore -\frac{d}{dt}(2\pi r) = 2v$$

$$\Rightarrow \frac{dr}{dt} = -\frac{v}{\pi}$$

$$\phi = B\pi r^2$$

$$\varepsilon = \left| -\frac{d\phi}{dt} \right|$$

$$\varepsilon = B2\pi r \left| \frac{dr}{dt} \right| = B2\pi r \frac{v}{\pi} = 2Brv$$

$$\Rightarrow 1 = \frac{\varepsilon}{R} = \frac{2Brv}{R}$$

$$l = \frac{2Brv}{\lambda(2\pi r)}$$

$$l = \frac{Bv}{\pi\lambda}$$

5. **Ans (C)**

$$\omega = \sqrt{\frac{k}{m}}$$

At equilibrium $k(l - a) = mg$

$$\Rightarrow \frac{k}{m} = \frac{g}{(l - a)}$$

$$\Rightarrow \text{Angular frequency, } \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{(l - a)}}$$

6. **Ans (B)**

$$\Delta r = (\mu_A - 1)t_A - (\mu_B - 1)t_B$$

$$= \mu_A t_A - t_A - \mu_B t_B + t_B$$

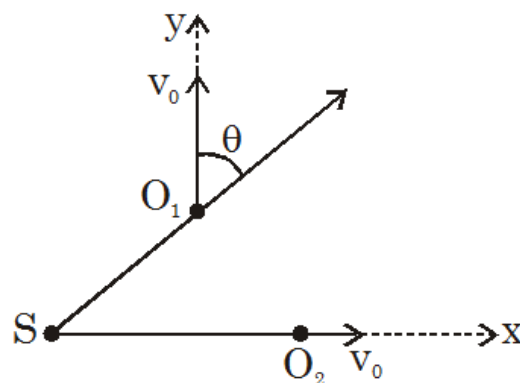
$$= t_B - t_A$$

$\Rightarrow$ If $\Delta r > 0$, then fringe pattern will shift upwards (towards A)

$\Rightarrow$ If $\Delta r < 0$, then fringe pattern will shift upwards (towards B)

7. **Ans (D)**

$$f_1 = f_0 \left(\frac{v - v_0 \cos \theta}{v} \right)$$



$$f_2 = f_0 \left(\frac{v - v_0}{v} \right)$$

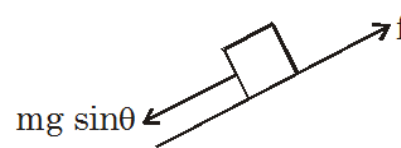
$$f_1 > f_2 \quad (\because v_0 \cos \theta < v_0)$$

$$\Rightarrow \left(\frac{f_1}{f_2} \right) > 1$$

8. **Ans (A)**

$$mg \sin \theta = \eta A \frac{v}{t}$$

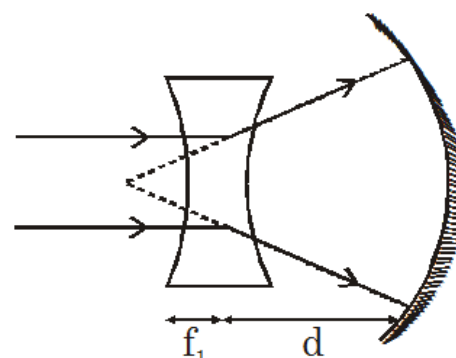
$$a^3 \rho g \times \frac{3}{5} = \eta \times a^2 \times \frac{v}{t}$$



$$\frac{3}{5} a \rho g = \frac{\eta v}{t}$$

$$\eta = \frac{3 \rho a g t}{5 v}$$

9. **Ans (B)**



$$(d + |f_1|) = 2|f_2|$$

$$\Rightarrow d = 2|f_2| - |f_1|$$

10. **Ans (B)**

As charge on the capacitor is conserved

$$\Rightarrow C_0 V_0 = C' V$$

$$\Rightarrow \frac{\epsilon_0 A}{d_0} V_0 = \frac{\epsilon_0 A}{d} V$$

$$\Rightarrow \frac{V_0}{d_0} = \frac{V}{d} \left(1 + \frac{1}{2} \sin \omega t \right)$$

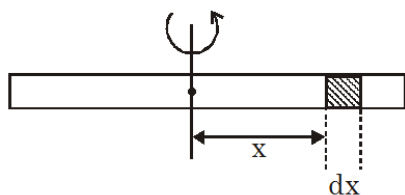
$$d_1 = \frac{d_0}{2}$$

11. **Ans (A)**

$$\text{Current, } I = \left(\frac{V}{\rho d/S} \right) = \frac{V S}{\rho d}$$

12. **Ans (A)**

$$\begin{aligned} dM &= IA = (dq) f \pi x^2 \\ &= \frac{q}{\ell} dx f \pi x^2 \end{aligned}$$



$$M = \frac{q f \pi}{\ell} \int_{-\ell/2}^{\ell/2} x^2 dx = \frac{q \pi f \ell^2}{12}$$

13. **Ans (D)**

$\vec{E}$ is parallel to $\vec{B}$. $\vec{v}_0$ is perpendicular to both.

Therefore path of the particle is a helix with increasing pitch. Speed of particle at any time t

$$\text{is } v = \sqrt{v_x^2 + v_y^2 + v_z^2} \dots (i)$$

$$\text{Here, } v_y^2 + v_z^2 = v_0^2$$

$$v = 2v_0$$

putting this in equation (i)

$$v_x = \sqrt{3}v \text{ and } v_x = \frac{qE_0}{m} t$$

$$t = \frac{\sqrt{3}mv_0}{qE_0}$$

14. **Ans (A)**

$$I = \frac{24}{\frac{RG}{R+G} + 1}$$

$$V = 24 - \text{potential difference across } 1\Omega = 24 - 1$$

$$V = 24 - \left(\frac{24}{\frac{1}{\frac{1}{R} + \frac{1}{G}} + 1} \right)$$

$$\text{For } G \rightarrow \infty \Rightarrow \frac{1}{G} \rightarrow 0 \text{ and } V = 20 \text{ volts}$$

$$\Rightarrow 20 = 24 - \frac{24}{R+1} \Rightarrow R = 5\Omega$$

15. **Ans (B)**

$$\begin{aligned} A &= \frac{\frac{d}{dt} \left[\int \vec{F} \cdot d\vec{s} \right]}{[\vec{F} \cdot \vec{P}]} \\ &= \frac{[\text{Work}]}{\text{time} \times \frac{\text{work}}{\text{displacement}} \times P} = \frac{1}{M} \end{aligned}$$

16. **Ans (A)**

Using conservation of momentum :

$$m(v_r - v) = 6mv$$

$$\Rightarrow v_r = 7v \dots (i)$$

Now, using conservation of energy:

$$mg\ell = \frac{1}{2} m(6v)^2 + \frac{1}{2} 6mv^2$$

$$\therefore v_r = 7v = \sqrt{\frac{7g\ell}{3}}$$

17. **Ans (C)**

After each collision normal components of velocity gets reversed.

$\therefore$ after 1st collision, velocity of coin,

$$\vec{v}_1 = 2\hat{i} - 3\hat{j}$$

after 2nd collision, velocity of coin,

$$\vec{v}_2 = -2\hat{i} - 3\hat{j}$$

after 3rd collision, velocity of coin,

$$\vec{v}_3 = -2\hat{i} + 3\hat{j}$$

18. **Ans (B)**

$$E_1 = \frac{KP_1}{r^3}$$

$$F = P_2 \left| \frac{dE_1}{dr} \right| = \frac{3KP_1 P_2}{r^4}$$

19. **Ans (D)**

Acceleration of centre of mass of the system,

$$a_c = \frac{2QE}{3m}$$

$$QE \left(\frac{2L}{3} \right) - QE \left(\frac{L}{3} \right) = \left(m \times \frac{4L^2}{9} + 2m \times \frac{L^2}{9} \right) \alpha$$

$$\frac{QEL}{3} = \frac{2mL^2}{3} \alpha$$

$$\alpha = \frac{QE}{2mL} \text{ clockwise}$$

$$a_A = a_C + \alpha \left(\frac{2L}{3} \right) = \frac{2QE}{3m} + \frac{QE}{2mL} \left(\frac{2L}{3} \right) = \frac{QE}{m}$$

20. **Ans (D)**

As it passes twice the transmission axis in one rotation.

PART-1 : PHYSICS

SECTION-II

1. **Ans (6)**

Reyleigh's criterion for the angular $\Delta\theta$ of a circular aperture of diameter $D \gg \lambda$ gives

$$\Delta\theta = 1.22 \frac{\lambda}{D}$$

and for two objects at distance z separated by s

the angular separation is

$$\Delta\theta = \frac{s}{z} \text{ (provided } s \ll z \text{)}$$

Combining these two expressions gives

$$D = 1.22 \frac{\lambda z}{s}$$

5. **Ans (9)**

Image distance is 30 cm

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f}$$

$$\frac{1}{30} = \frac{2}{f}$$

$$f = 60 \text{ cm}$$

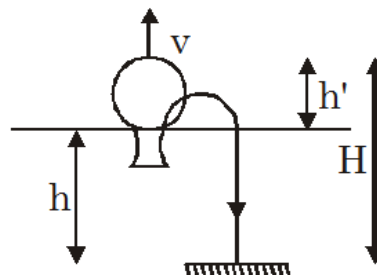
As optical axis shifts down by $(y - x)$ for second lens

$$m = \frac{y - x - 1}{y - x}$$

$$y = 4.5 \text{ cm}$$

6. **Ans (245)**

The height of the balloon, when the body is released, the body will have same initial velocity as the balloon.



$$S = -h = ut - \frac{1}{2}gt^2$$

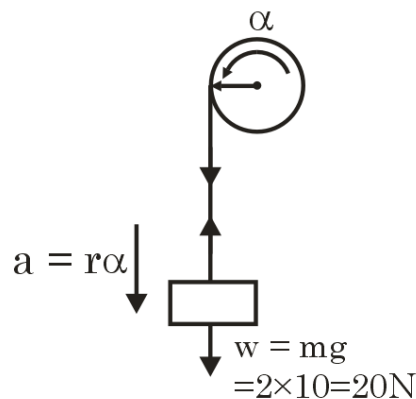
$$\therefore h = \frac{1}{2}gt^2 - ut$$

$$h = 233.6 \text{ m}$$

Above the balloon frame, the body will go up by,

$$h' = \frac{u^2}{2g} = \frac{100}{2 \times 9.8} = 5.10 \text{ m}$$

7. **Ans (2)**



$$Mg - T = Ma$$

$$TR = I\alpha$$

$$Mg - \frac{I\alpha}{R} = Ma$$

$$Mg = \left(MR + \frac{I}{R} \right) \alpha$$

$$\alpha = \frac{Mg}{MR + \frac{I}{R}}$$

$$\alpha = \frac{MgR}{MR^2 + I}$$

$$\alpha = \frac{2 \times 10 \times 1}{2(1) + 8}$$

$$\alpha = 2 \text{ rad/s}^2$$

8. Ans (2)

'g' at poles, $g = \frac{GM}{R^2} \rightarrow (1)$

At the equator, $mg - mg' = \frac{mv^2}{R}$

$mg - \frac{mg}{2} = \frac{mv^2}{R}$

$g = \frac{2V^2}{R} \rightarrow (1)$

Let v be the required escape velocity at the poles, then total energy of the particle is,

For the particle to escape its total energy must be at least equal to the minimum energy of a body at infinity i.e., zero

$\frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0$

$v_e^2 = \frac{2GM}{R} = 2gR$

From (1) :

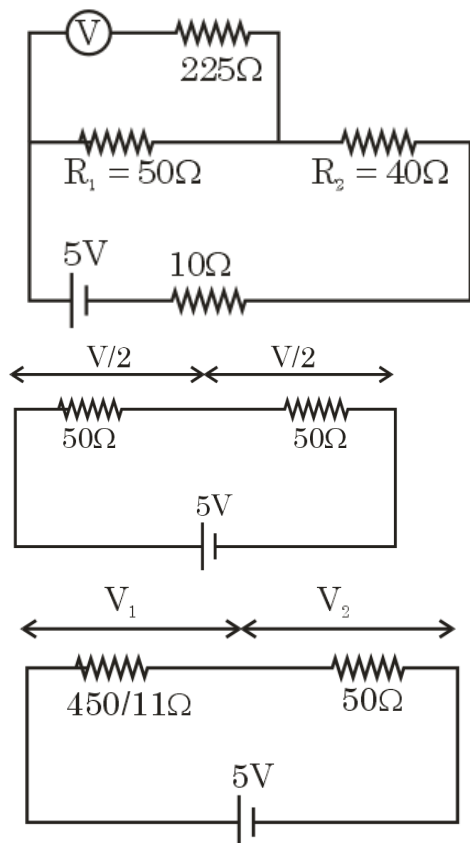
$v_e^2 = 2 \left(\frac{2V^2}{R} \right) R \quad \left(g = \frac{2V^2}{R} \right)$

$v_e^2 = 4V^2$

$\therefore v_e = 2V$

$\therefore k = 2$

9. Ans (2)



Error = $\frac{\frac{V}{2} - \frac{9V}{20} \times 100}{\frac{V}{2}} \quad V_1 : V_2 :: \frac{450}{11} : 50$

Error = 10% $V_1 : V_2 :: 9 : 11$

$V_1 = \frac{9}{20} V$

10. Ans (5)

Since, $I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$

Where $I = \frac{3}{4}(4I_0) = 3I_0$

$\Rightarrow \cos \left(\frac{\phi}{2} \right) = \pm \frac{\sqrt{3}}{2}$

$\Rightarrow \left(\frac{\phi}{2} \right) = n\pi \pm \frac{\pi}{6}$

$\Rightarrow \phi = 2n\pi \pm \frac{\pi}{3}$

Since, $\phi = \frac{2\pi}{\lambda} \Delta x$ where $\Delta x = \frac{y_n d}{D}$

$\Rightarrow y_n = \left(n \pm \frac{1}{6} \right) \frac{\lambda D}{d}$

For the point lying between third minima and third maxima, We have $n = 3$

$\Rightarrow y_3 = \left(3 - \frac{1}{6} \right) \frac{\lambda D}{d}$

$\Rightarrow y_3 = \frac{17}{6} \left(\frac{\lambda D}{d} \right)$

Substituting $\lambda = 0.6 \times 10^{-6} \text{ m}$, $D = 1/2 \text{ m}$, $d = 17 \text{ mm}$,

we get

$y_3 = \frac{17}{6} \frac{(0.6 \times 10^{-6}) (1/2)}{17 \times 10^{-3}} = 5 \mu\text{m}$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)

$\pi = i.CRT$

$i = \frac{8.21}{0.15 \times 0.0821 \times 300} = 2.22$

2. Ans (A)

Let solubility barium sulphite is x mol/L and that of barium sulphate is y mol/L.

$(x + y)x = 8 \times 10^{-7}$

$(x + y)y = 1 \times 10^{-8}$

3. **Ans (D)**

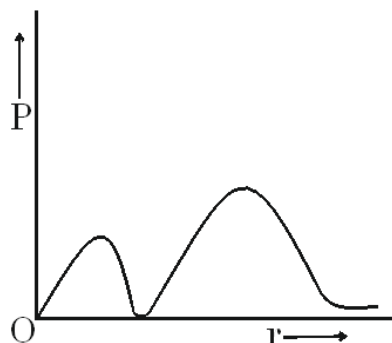
$\text{Fe}(\text{SCN})_3$ is red colour

4. **Ans (D)**

Brown ring complex is $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{+2}$

7. **Ans (D)**

Sketch for 2s orbital is



8. **Ans (B)**

Hemimorphite is pyrosilicate.

9. **Ans (A)**

P_4O_{10} is not an oxidizing agent

$\text{P}_4\text{O}_{10} + \text{NaCl} \rightarrow \text{POCl}_3 + \text{NaPO}_3$

10. **Ans (B)**

Barytes is a sulphate ore.

12. **Ans (B)**

It contains no symmetry element.

13. **Ans (D)**

(A) Brownian movement is one of the reason for stability of sols.

(B) Catalyst can undergo qualitative/physical change but cannot undergo any quantitative change.

(C) Both heating & freezing cause de-emulsification.

(D) Excessive dialysis causes co-agulation in colloids making them unstable.

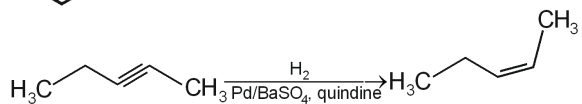
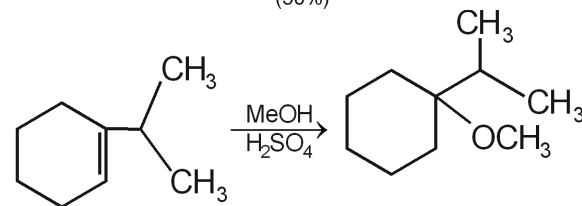
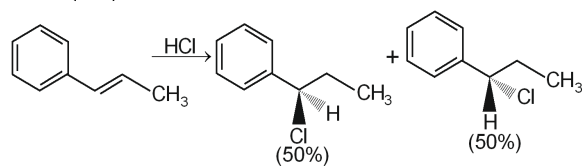
14. **Ans (C)**

$\text{B}_3\text{N}_3\text{H}_6 + 3\text{HCl} \rightarrow \text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$

15. **Ans (D)**

$\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\Delta} 2\text{NaBO}_2 + \text{B}_2\text{O}_3$

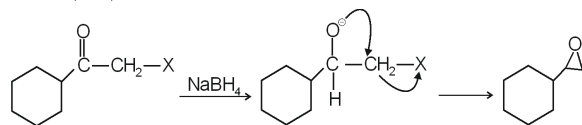
17. **Ans (D)**



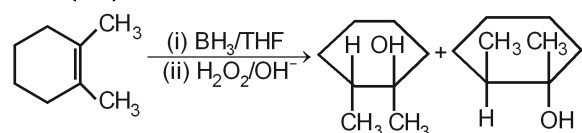
18. **Ans (B)**

Conjugate base formed after removal of H (b) is resonance stabilized by both keto and alkene group.

19. **Ans (B)**



20. **Ans (D)**



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (4)**

Numbers of mole of A = $10/150 = 1/15$

Moles of B = $0.20 \times (200/1000) = 0.04$

A is limiting reagent hence moles of X formed

= $(1/15)/2$

P = $(1/30) \times 0.08 \times 300/0.2 = 4.00 \text{ atm}$

2. **Ans (250)**

$$\text{Degree of dissociation } (\alpha) = \frac{\Lambda_m}{\Lambda_m^0}$$

$$\Rightarrow \Lambda_m = \Lambda_m^0 \times \alpha$$

$$= \Lambda_m^0 \times \sqrt{\frac{K_a}{C}} \text{ (by Ostwald dilution law)}$$

$$= 400 \times \sqrt{\frac{1.8 \times 10^{-5}}{0.18}} = 4.00 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\text{Conductivity} = \frac{\text{molarity} \times \Lambda_m}{1000}$$

$$= 7.2 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$\text{Also conductivity} = \text{Cell constant} \times \frac{1}{\text{Resistance}}$$

$$\text{Resistance} = \frac{\text{Cell constant}}{\text{Conductivity}}$$

$$= \frac{0.18}{7.2 \times 10^{-4}} = 284.9 \text{ ohm}$$

3. **Ans (200)**

Let the final composition, liquid (10 mole)

A = x mole, B = (10 - x) mole

Vapour (10 moles), A = 10 - x mole; B = x mole

$$\text{Now, } \frac{Y_A}{Y_B} = \frac{X_A}{X_B} \frac{P_A^0}{P_B^0} \Rightarrow \frac{10-x}{x} = \frac{x}{10-x} \times \frac{400}{100}$$

$$x = \frac{10}{3}$$

$$\text{Now, } P = X_A P_A^0 + X_B P_B^0$$

$$= \frac{x}{10} \times 400 + \frac{10-x}{10} \times 100$$

$$= 200.00$$

4. **Ans (85)**



n-factor of $\text{K}_2\text{Cr}_2\text{O}_7 = 6$ and

$$\text{n-factor of HCl} = \frac{3}{7}$$

$$\therefore E_{\text{HCl}} = \frac{M}{3} = \frac{7M}{3} = 85.17$$

5. **Ans (7)**

$$\Delta H = nC_p \Delta T = 2 \times \frac{7}{2} R \times (1500 - 1000)$$

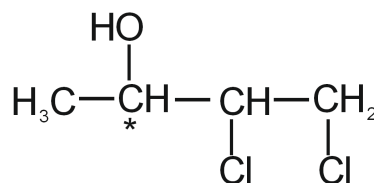
$$= 2 \times \frac{7}{2} \times 2 \times 500 = 7000 \text{ cal}$$

6. **Ans (58)**

Y is allylic alcohol

7. **Ans (4)**

Two chiral center so 4 stereo isomers are possible for the given organic molecule



8. **Ans (10)**

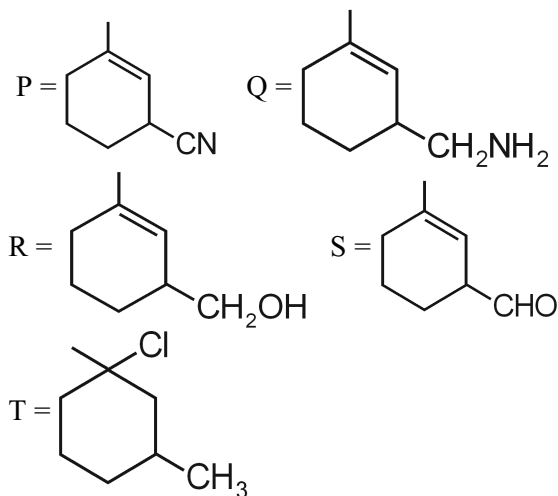
Total 5 dihalogen derivatives are formed including stereoisomers in which two are diastereomers.

$$\therefore x + y = 10$$

9. **Ans (4)**

there are 4 double bonds so 4 moles of H_2 needed

10. **Ans (147)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

$$\text{Let } a = \sqrt{\frac{x+y+z}{xyz}}$$

$$\tan^{-1}(ax) + \tan^{-1}(ay) + \tan^{-1}(az)$$

$$= \pi + \tan^{-1} \left(\frac{ax + ay + az - a^3xyz}{1 - (\dots)} \right);$$

$$(\text{where } a^2(xy + yz + zx) > 1)$$

$$= \pi + \tan^{-1} \left(\frac{a(x+y+z - a^2xyz)}{dr} \right)$$

$$= \pi + \tan^{-1} \left(\frac{0}{dr} \right) = \pi$$

2. **Ans (B)**

$$\text{Line } L : \frac{x-1}{1} = \frac{y-2}{-5} = \frac{z-1}{3} = \lambda(\text{say})$$

$$Q = (\lambda + 1, 2 - 5\lambda, 3\lambda + 1)$$

$$\vec{PQ} = (\lambda, -5\lambda, 3\lambda)$$

$$\Rightarrow |\vec{PQ}|^2 = 35$$

$$\Rightarrow \lambda = \pm 1$$

$$Q = (\lambda + 1, 2 - 5\lambda, 3\lambda + 1) \begin{cases} (2, -3, 4) \\ (0, 7, -2) \end{cases}$$

$$\text{then } a + b + c \begin{cases} (2, -3, 4) \Rightarrow 3 \\ (0, 7, -2) \Rightarrow 5 \end{cases}$$

3. **Ans (B)**

$$B = A_1 + 3A_3 + 5A_5 + \dots + (2n-1)A_{2n-1}^{2n-1}$$

taking transpose both sides, then we get

$$\Rightarrow B^T = -B$$

$\Rightarrow B$ is skew symmetric matrix

4. **Ans (B)**

$$\sum_{i=0}^n i \cdot {}^nC_i \cdot p^i(1-p)^{n-i}$$

$$= \sum_{i=1}^n n \cdot {}^{n-1}C_{i-1} \cdot p^{i-1} \cdot p(1-p)^{n-i}$$

$$= np \sum_{i=1}^n {}^{n-1}C_{i-1} \cdot p^{i-1}(1-p)^{n-i}$$

$$= np(1) = np$$

5. **Ans (A)**

$$12 \begin{cases} 7G \\ 5R \end{cases}$$

$\Rightarrow P(\text{other mango is also good})$

$$= \frac{{}^7C_2}{{}^7C_2 + {}^7C_1 \cdot {}^5C_1}$$

$$= \frac{3}{8}$$

6. **Ans (B)**

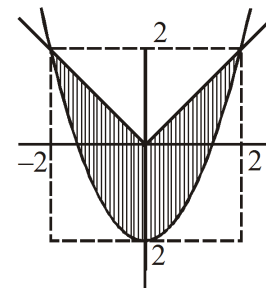
In $I_1 \rightarrow$ Apply king property and add

$$\Rightarrow 2I_1 = 2 + 2I_2$$

$$\Rightarrow I_1 - I_2 = 1$$

7. **Ans (B)**

$$\text{Required area} = \frac{2}{3} \times 4 \times 4 - \frac{1}{2} \times 4 \times 2 = \frac{20}{3}$$



8. **Ans (A)**

$$(1 + \ln x) \frac{dx}{dy} - x \ln x = e^y$$

$$\left. \begin{aligned} \text{substitute } x \ln x &= t \\ (1 + \ln x) \frac{dx}{dy} &= \frac{dt}{dy} \end{aligned} \right\}$$

$$\Rightarrow \frac{dt}{dy} - t = e^y$$

$$\Rightarrow \text{I.F.} = e^{-\int dy} = e^{-y}$$

$$\Rightarrow \text{Sol. } te^{-y} = \int e^{-y} \cdot e^y dy + c$$

$$\Rightarrow t = ye^y + ce^y$$

$$\Rightarrow x \ln x = ye^y + ce^y, \text{ put } x = 1, y = 0$$

$$0 = c, \text{ then we get}$$

$$\Rightarrow x \ln x = ye^y$$

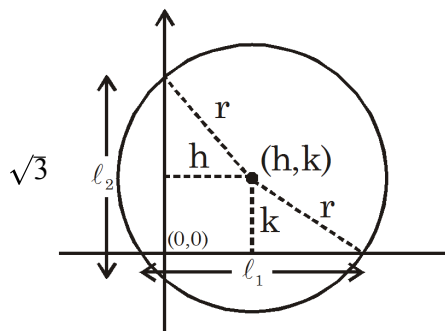
$$\ln x^x = y e^y$$

$$x^x = e^{ye^y}$$

9. **Ans (B)**

$$k = \left| \frac{\frac{1}{2} (\vec{a} \times (2\vec{a} + 3\vec{b}) + (2\vec{a} + 3\vec{b}) \times \vec{b})}{\vec{a} \times \vec{b}} \right| = \frac{5}{2}$$

10. Ans (C)



$$r^2 = \frac{l_1^2}{4} + k^2 = \frac{l_2^2}{4} + h^2$$

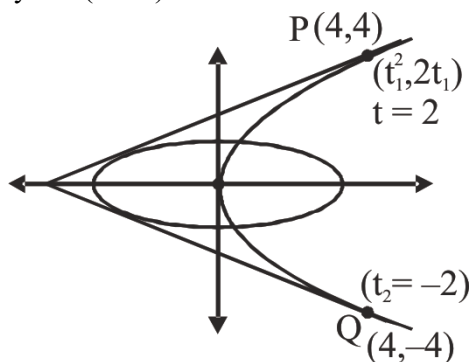
$$h^2 - k^2 = \frac{l_1^2 - l_2^2}{4}$$

$\Rightarrow$ which is rectangular hyperbola whose eccentricity is $\sqrt{2}$

11. Ans (A)

Equation of tangent to the parabola $y^2 = 4x$, at $Q(4, -4)$ is :-

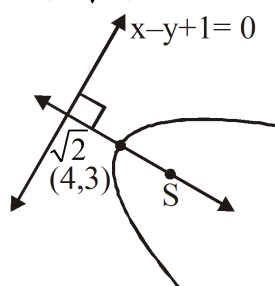
$$\Rightarrow -4y = 2(x + 4)$$



12. Ans (B)

$$\tan \theta = -1,$$

$$s \equiv \left(4 - \sqrt{2} \left(-\frac{1}{\sqrt{2}} \right), 3 - \sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right)$$



$$s \equiv (5, 2)$$

13. Ans (A)

$$\frac{dv}{dt} = 2 = \frac{d}{dt} \left(\frac{\pi r^3}{3} \right) \quad (h = r, \theta = 45^\circ)$$

$$\frac{dr}{dt} = \frac{2}{\pi r^2}$$

$$P = 2\pi r$$

$$\frac{dp}{dt} = 2\pi \frac{dr}{dt} = 2\pi \times \frac{2}{\pi r^2} = \frac{4}{r^2}$$

$$\left(\frac{dp}{dt} \right)_{r=2} = \frac{4}{4} = 1 \text{ cm/sec}$$

14. Ans (B)

$$\frac{x^2}{9} + \frac{y^2}{9-25} = 1$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$e = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}$$

15. Ans (D)

In dual statement $\vee$ replace by $\wedge$ and $\wedge$ replace by $\vee$

so answer is $(p \wedge \sim q) \vee (\sim p)$

16. Ans (D)

$$\Rightarrow \frac{3!}{2! \cdot 1!} \cdot {}^3C_1 \cdot {}^3C_1 \cdot {}^3C_2 \cdot \frac{6!}{2! \cdot 2! \cdot 2!}$$

$$= 81 \times 90 = 7290$$

17. Ans (D)

R.H.L. =

$$\lim_{x \rightarrow 0^+} \frac{x}{\tan 7x} \cdot \lim_{x \rightarrow 0^+} (8x+9)^{\frac{1}{\pi} \tan^{-1} \frac{1}{x}} \cdot \lim_{x \rightarrow 0^+} \frac{x + \sin 6x}{\int_0^x e^{t^2} dt}$$

$$= \frac{1}{7} \times (9)^{\frac{1}{\pi} \cdot \frac{\pi}{2}} \cdot \frac{1 + 6 \cos 6x}{e^{x^2}}$$

(Use L'Hopital rule in third part)

$$= \frac{1}{7} \times (9)^{1/2} \cdot \left(\frac{1+6}{1} \right) = 3$$

L.H.L. =

$$\lim_{x \rightarrow 0^-} \frac{x}{\tan 7x} \cdot \lim_{x \rightarrow 0^-} (8x+9)^{\frac{1}{\pi} \tan^{-1} \frac{1}{x}} \cdot \lim_{x \rightarrow 0^-} \frac{x + \sin 6x}{\int_0^x e^{t^2} dt}$$

$$= \frac{1}{7} \times (9)^{-1/2} \cdot 7 = \frac{1}{3}$$

L.H.L. $\neq$ R.H.L.

Hence limit does not exist.

18. Ans (D)

$$f(x) = 1,$$

$$\Rightarrow \int \underbrace{\frac{x}{\sqrt{x^2+1}}}_{II} \cdot \underbrace{\ln(x + \sqrt{1+x^2})}_I dx$$

Apply by parts

$$\Rightarrow \sqrt{x^2+1} \cdot \ln(x + \sqrt{1+x^2}) - \int 1 \cdot dx$$

$$\Rightarrow \sqrt{x^2+1} \cdot \ln(x + \sqrt{1+x^2}) - x + c$$

19. Ans (A)

$$\bar{x}_1 = 520, \bar{x}_2 = 420, \bar{x} = 500$$

Let n_1 is number of male employees

n_2 is number of female employees

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$500(n_1 + n_2) = 520n_1 + 420n_2$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{4}{1}$$

% male employees

$$\Rightarrow \frac{4}{4+1} \times 100 = 80\%$$

20. Ans (C)

$$y = f(t), -3 \leq t \leq 3 \text{ and } -9 \leq 3y \leq 15$$

$$-3 \leq 2x + 1 \leq 3 \quad -7 \leq (2 + 3y) \leq 17$$

$$-2 \leq x < 1 \quad -7 \leq g(x) \leq 17$$

$$a = -2, b = 1 \quad c = -7, d = 17,$$

$$\text{then } a + b + c + d = -2 + 1 + 17 - 7 = 9$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1)

$$f'(x) = 3x^2 + 2x + 3 + 2 \cos x$$

$$= \underbrace{3\left(x + \frac{1}{3}\right)^2}_{\geq 0} - \underbrace{\frac{1}{3} + 3 + 2 \cos x}_{> 0},$$

$f(x)$ is strictly increasing and $f(0) = 0$

2. Ans (252)



$$f(2) < 0 \Rightarrow 12 - 24 + \frac{\lambda}{3} < 0$$

$$\Rightarrow \lambda < 36 \quad \dots(i)$$

$$f(1) > 0 \Rightarrow 3 - 12 + \frac{\lambda}{3} > 0$$

$$\Rightarrow \lambda > 27 \quad \dots(ii)$$

$$f(3) > 0 \Rightarrow 27 - 36 + \frac{\lambda}{3} > 0$$

$$\Rightarrow \lambda > 27 \quad \dots(iii)$$

eq. (i) $\cap$ (ii) $\cap$ (iii), then we get

$$\lambda \in (27, 36)$$

$$\lambda = 28, 29, 30, 31, 32, 33, 34, 35$$

$$\lambda = 28 + 29 + 30 + 31 + 32 + 33 + 34 + 35 = 252$$

3. Ans (3)

$$|\vec{c} - \vec{a}|^2 = 8$$

$$\Rightarrow (|\vec{c}| - 1)^2 = 0$$

$$\Rightarrow |\vec{c}| = 1 \text{ and } |\vec{a} \times \vec{b}| = 3,$$

$$\Rightarrow \left(|\vec{a} \times \vec{b}| |\vec{c}| \cos \phi \right)_{\max},$$

when $\cos \phi = 1$

$$\text{then } \left(|\vec{a} \times \vec{b}| |\vec{c}| \cos \phi \right)_{\max} = 3$$

4. Ans (33)

$$S = \frac{3}{11} + \frac{33}{11^2} + \frac{333}{11^3} + \dots$$

$$\frac{S}{11} = \frac{3}{11^2} + \frac{33}{11^3} + \dots$$

$$\frac{10}{11} S = \frac{3}{11} + \frac{30}{11^2} + \frac{300}{11^3} + \dots$$

$$S = \frac{11}{10} \times \frac{\frac{3}{11}}{1 - \frac{10}{11}} = \frac{33}{10}$$

5. Ans (0)

$$b = a^2 \text{ and } s = \frac{-8}{r}$$

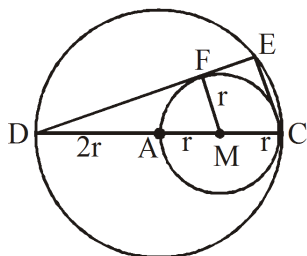
$$\frac{s - b}{r - a} = 2a = \frac{8}{r^2}$$

$$\Rightarrow a = 4 \text{ and } r = 1,$$

$$b = 16, s = -8$$

6. **Ans (8)**

$$\frac{DE}{DF} = \frac{DC}{DM} = \frac{4}{3}$$



$$DE = \frac{4}{3} \sqrt{(3r)^2 - r^2} = \frac{8\sqrt{2}r}{3}$$

$$= \frac{8\sqrt{2} \times 6\sqrt{2}}{3 \times 4} = 8$$

7. **Ans (25)**

$$a_i = \frac{20}{i} \Rightarrow \frac{20 \times 20 \times 20}{6 \times 20 \times 8} \begin{vmatrix} 6 & 2 & 3 \\ 5 & 1 & 4 \\ 4 & 1 & 2 \end{vmatrix}$$

$$= \frac{50}{6} \times 3 = 25$$

8. **Ans (19)**

$$a^2 + 1 = 2 \Rightarrow a = 1$$

$$5 = c - \frac{b^2}{3} \text{ and } 1 = \frac{b^2}{9}$$

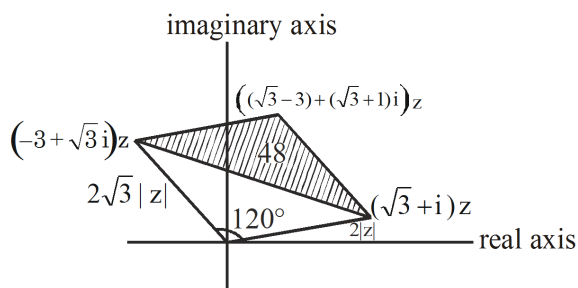
$$\Rightarrow b = 3, c = 8$$

9. **Ans (4)**

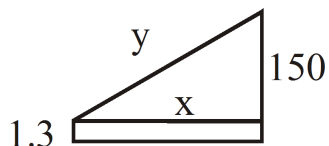
$$\text{Area} = \frac{1}{2} \times 2 |z| \times 2\sqrt{3} |z| \sin 120^\circ = 48 \text{ (given)}$$

$$|z|^2 \times 2\sqrt{3} \times \frac{\sqrt{3}}{2} = 48$$

$$|z| = 4$$



10. **Ans (8)**



$$y^2 = x^2 + 150 ; \text{ given that } \frac{dx}{dt} = 10$$

$$\frac{dy}{dt} = \frac{x}{y} \times \frac{dx}{dt}$$

$$\text{at } y = 250, x = 200$$

$$\frac{dy}{dx} = \frac{200}{250} \times 10 = 8 \text{ m/s}$$